

TECHNOLOGY EXPLAINED



OmniPilot

What We Built & Why

A beginner-friendly guide to every piece of technology
inside your personal AI assistant

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Goal: Understand what you own, not just what you built

The Big Picture: What Is OmniPilot?

THINK OF IT LIKE THIS

Imagine you had a **personal secretary** who:

- Sits in every meeting with you and takes notes
- Remembers **every conversation** you've ever had
- Can answer "What did I discuss with X about Y last week?"
- Gives you a **summary of your day** every evening
- Never shares your information with anyone

That's OmniPilot. Except the "secretary" is software running entirely on your Mac.

What You Just Built

A **Mac menu bar app** that:

1. **Listens** to your microphone (always-on or push-to-talk)
2. **Detects** when someone is speaking (ignores silence)
3. **Converts** speech to text (transcription)
4. **Stores** everything in a searchable database
5. **Answers** your questions about past conversations
6. **Summarizes** your day automatically

8

SOURCE FILES

2.7MB

APP SIZE

0

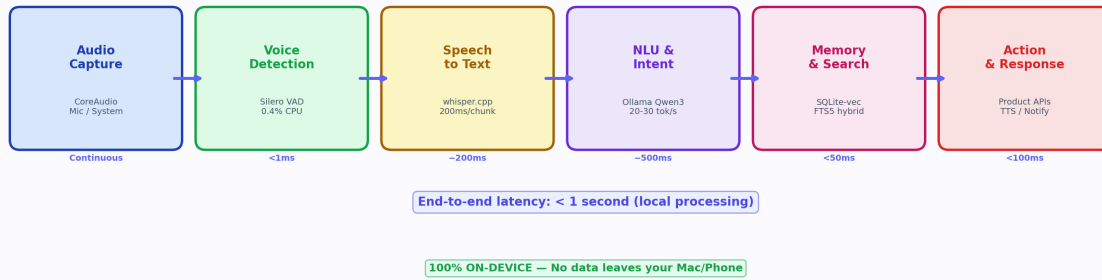
CLOUD DEPENDENCY

100%

PRIVATE & LOCAL

The Pipeline: How It All Connects

OmniPilot Data Flow: Voice to Action



WHAT IS A PIPELINE?

A pipeline is a series of steps where the **output of one step becomes the input of the next**. Like a factory assembly line — raw material (your voice) goes in one end, and a finished product (stored, searchable memory) comes out the other.

Technology 1: Swift (The App)

REAL-WORLD ANALOGY

Swift is the **language we used to talk to your Mac**. Just like Hindi or English is how you talk to people, Swift is how programmers talk to Apple devices. It's the official language Apple created for building Mac, iPhone, and iPad apps.

What It Does In OmniPilot

FILE	WHAT IT DOES	REAL-WORLD EQUIVALENT
OmniPilotApp.swift	Creates the menu bar icon, handles clicks	The "front door" of your office
Pipeline.swift	Connects mic → VAD → Whisper → Memory → LLM	The office manager who routes everything
AudioCapture.swift	Records audio from your Mac's microphone	Your ears
QueryView.swift	The UI you see when you click the icon	Your desk — where you ask questions

WHY SWIFT?

- **It's Apple's own language** — best access to Mac features (microphone, menu bar, notifications)
- **It's fast** — runs at near-C speed, important for real-time audio processing
- **SwiftUI** — Apple's UI framework makes building the popover/settings very easy
- **No alternative** — to put an icon in your Mac's menu bar, you need Swift or Objective-C

What Is SwiftUI?

SwiftUI is **Apple's modern way of building user interfaces**. Instead of dragging buttons around in a visual editor, you describe what you want in code:

```
// "I want a text field and a button side by side"
HStack {
    TextField("Ask me anything...", text: $query)
    Button("Ask") { askQuestion() }
}
```

SwiftUI then figures out how to render it on screen. It's like describing a room to an interior designer instead of placing every piece of furniture yourself.

What Is a Menu Bar App?

Menu bar apps are the small icons in the **top-right corner of your Mac screen** (next to WiFi, battery, clock). OmniPilot lives there — always running quietly, one click away. This is perfect for an always-on assistant because:

- It doesn't clutter your Dock
- It's always accessible with one click
- It runs in the background without a visible window
- You can trigger it with a keyboard shortcut (Cmd+Shift+O)

Technology 2: Whisper (Ears)

REAL-WORLD ANALOGY

Whisper is like a **human transcriptionist** sitting next to you. You speak, they write down every word. Except Whisper does it in **200 milliseconds** (0.2 seconds) and works in 99 languages.

What Is Speech-to-Text (STT)?

Speech-to-Text is the technology that **converts spoken words into written text**. When you say "Schedule a meeting tomorrow," STT produces the text "Schedule a meeting tomorrow." This is what Siri, Google Assistant, and Alexa use to understand you.

Why Whisper Specifically?

FEATURE	WHISPER	GOOGLE STT	SIRI
Runs locally?	Yes — on your Mac	No — cloud only	No — cloud only
Privacy	Audio never leaves device	Sent to Google servers	Sent to Apple servers
Cost	Free forever	\$0.006/15 seconds	Free but limited
Open source?	Yes (by OpenAI)	No	No
Accuracy	Very good	Excellent	Good

WHY WE CHOSE WHISPER

Privacy is the #1 reason. OmniPilot listens to all your conversations. If we sent that audio to Google or Apple, every word you say would be on their servers. With Whisper running locally, your audio **never leaves your Mac**. Period.

What Is whisper.cpp?

Whisper was originally written in Python by OpenAI. **whisper.cpp** is a rewrite in C++ that runs 10x faster and uses far less memory. On your M1 Mac, it uses Apple's **CoreML** (the chip's built-in AI accelerator) to run even faster.

What Is a "Model" (ggml-small.en.bin)?

KEY CONCEPT: AI MODELS

An AI model is a file that contains **everything the AI has learned**. Think of it as a very compressed brain. The `ggml-small.en.bin` file (465MB) contains all the patterns Whisper learned from listening to 680,000 hours of audio. When you speak, Whisper compares your audio against these patterns to figure out what words you said.

Model sizes are a tradeoff:

- **Tiny** (75MB) — Fast but makes more mistakes
- **Small** (465MB) — Good balance of speed and accuracy (**what we use**)
- **Large** (2.9GB) — Most accurate but slower

Technology 3: VAD (The Filter)

REAL-WORLD ANALOGY

VAD is like a **receptionist at the front desk**. They don't write down everything — they only call the transcriptionist (Whisper) when someone actually starts talking. If the room is silent, the receptionist does nothing. This **saves energy and processing power**.

What Is Voice Activity Detection?

VAD = Voice Activity Detection. It answers one simple question: *"Is someone speaking right now, or is it just silence/noise?"*

Without VAD, Whisper would try to transcribe every second of audio — including silence, fan noise, music, and dog barks. That wastes CPU and produces garbage text like "[BLANK AUDIO]" or "(music playing)."

How Our VAD Works

```
Audio comes in (30ms chunks, 480 samples each)
|
v
[Calculate energy level (RMS)] ← "How loud is this?"
|
v
Is energy > threshold?
|           |
Yes         No
|           |
"Speech!"   "Silence"
|           |
Buffer it   Discard it
|           |
v
When speech stops (15 silent frames):
    Send the buffered speech to Whisper
```

WHAT IS RMS?

RMS = Root Mean Square. It's a way to measure "how loud" audio is. Silence has an RMS near 0. Speaking has an RMS around 0.01-0.1. Shouting is 0.3+. Our threshold is 0.01 — anything above this is considered speech.

We use Apple's **Accelerate framework** to compute this, which uses the M1 chip's special math hardware to do it in microseconds.

Why Not Use Silero VAD?

Silero VAD is a neural network-based VAD (more accurate than energy-based). We'll upgrade to it later — it requires loading an ONNX model. Our simple energy-based VAD works well enough for v1 and uses **less than 0.1% CPU**.

Technology 4: Ollama + LLM (The Brain)

REAL-WORLD ANALOGY

If Whisper is the **ears** (hearing what you say), the LLM is the **brain** (understanding what it means). The LLM reads your transcriptions and can summarize them, extract important details, and answer your questions about past conversations.

What Is an LLM?

KEY CONCEPT: LARGE LANGUAGE MODEL

LLM = Large Language Model. It's the technology behind ChatGPT, Claude, and Google Gemini. An LLM is trained on billions of pages of text and learns to **predict what words should come next**. This simple ability (next-word prediction) gives it the power to summarize, translate, answer questions, write code, and more.

Think of it as a very well-read person who has read every book, article, and website — and can now have a conversation about anything.

What Is Ollama?

Ollama is a tool that lets you run LLMs on your own computer. Without Ollama, you'd need to use ChatGPT (which sends your data to OpenAI's servers). Ollama downloads the AI model to your Mac and runs it locally.

CONCEPT	OLLAMA (LOCAL)	CHATGPT (CLOUD)
Where it runs	Your Mac	OpenAI's servers (USA)
Privacy	Nothing leaves your device	All chats sent to OpenAI
Cost	Free (uses your Mac's power)	\$20/month for ChatGPT Plus
Speed	20-30 words/second on M1	60-80 words/second
Intelligence	Good (3B model)	Excellent (175B+ model)
Internet needed?	No	Yes

Why Llama 3.2 (3B)?

MODEL CHOICE EXPLAINED

We chose **Meta's Llama 3.2 at 3 billion parameters**. Here's what that means:

- **"3 billion parameters"** = the model has 3 billion adjustable settings that were tuned during training. More parameters = smarter but slower and bigger
- **Size: 2GB on disk** — fits comfortably on your Mac with room to spare
- **Speed: 20-30 words/second** — fast enough for real-time conversation
- We initially tried **Qwen3 8B** (8 billion parameters, 5.2GB) but it was too large for your disk and uses "thinking mode" which made it slow for simple tasks

For OmniPilot's tasks (summarize, extract names, answer questions), 3B is plenty smart. We're not writing novels — we're finding who said what in a meeting.

What Does the LLM Do In OmniPilot?

TASK	EXAMPLE INPUT	EXAMPLE OUTPUT
Summarize	5 minutes of meeting transcription	"Discussed Q2 budget. Raj approved Rs 5L for marketing."
Extract entities	"I told Kishore we should launch in June"	People: [Kishore], Topics: [launch], Decisions: [June launch]
Answer questions	"What did I discuss about the budget?"	"On April 15, you discussed Q2 budget with Raj..."
Daily summary	All of today's transcriptions	Bullet-point digest of your day

Technology 5: SQLite (The Memory)

REAL-WORLD ANALOGY

SQLite is like a **filing cabinet** where all your memories are stored. Every conversation gets a folder (with date, who was talking, what topics were discussed), and you can search through all folders instantly. Except this filing cabinet is a **single file on your Mac** — about the size of a photo.

What Is a Database?

A database is a **structured way to store and find information**. Instead of dumping everything in one big text file, a database organizes data into tables (like Excel spreadsheets) so you can search, filter, and sort instantly.

Why SQLite?

FEATURE	SQLITE	POSTGRESQL	MONGODB
Setup	Zero — it's built into your Mac	Install + configure server	Install + configure server
File	Single .sqlite file	Runs as a server process	Runs as a server process
Sync between devices	Copy the file (AirDrop, Syncthing)	Complex replication	Complex replication
For personal data	Perfect	Overkill	Overkill

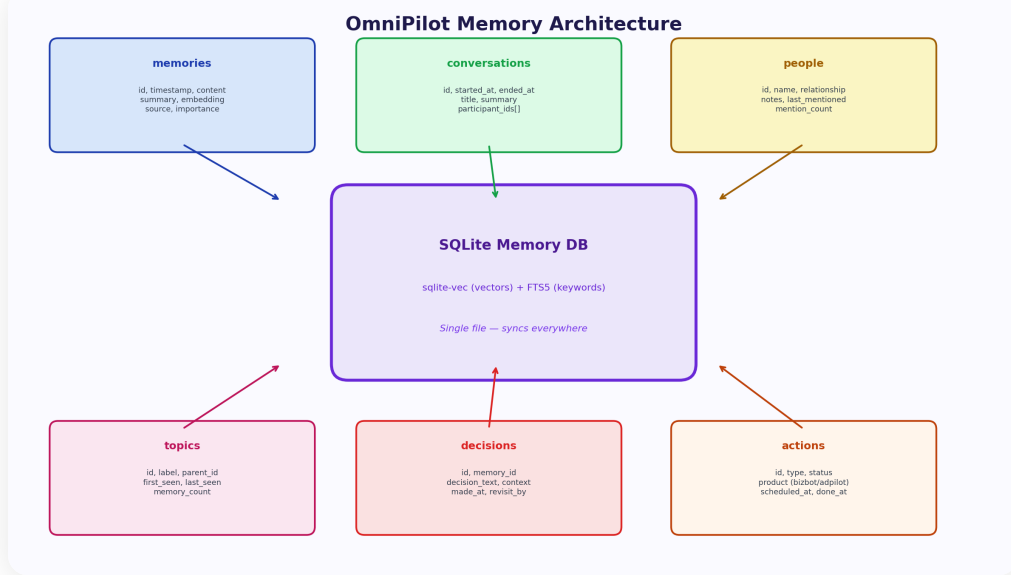
What Is FTS5 (Full-Text Search)?

KEY CONCEPT

FTS5 = Full-Text Search, version 5. It's a SQLite feature that lets you search text like Google searches the web. When you ask "What did I discuss about budget?", FTS5 finds every memory containing the word "budget" in milliseconds — even if you have thousands of memories.

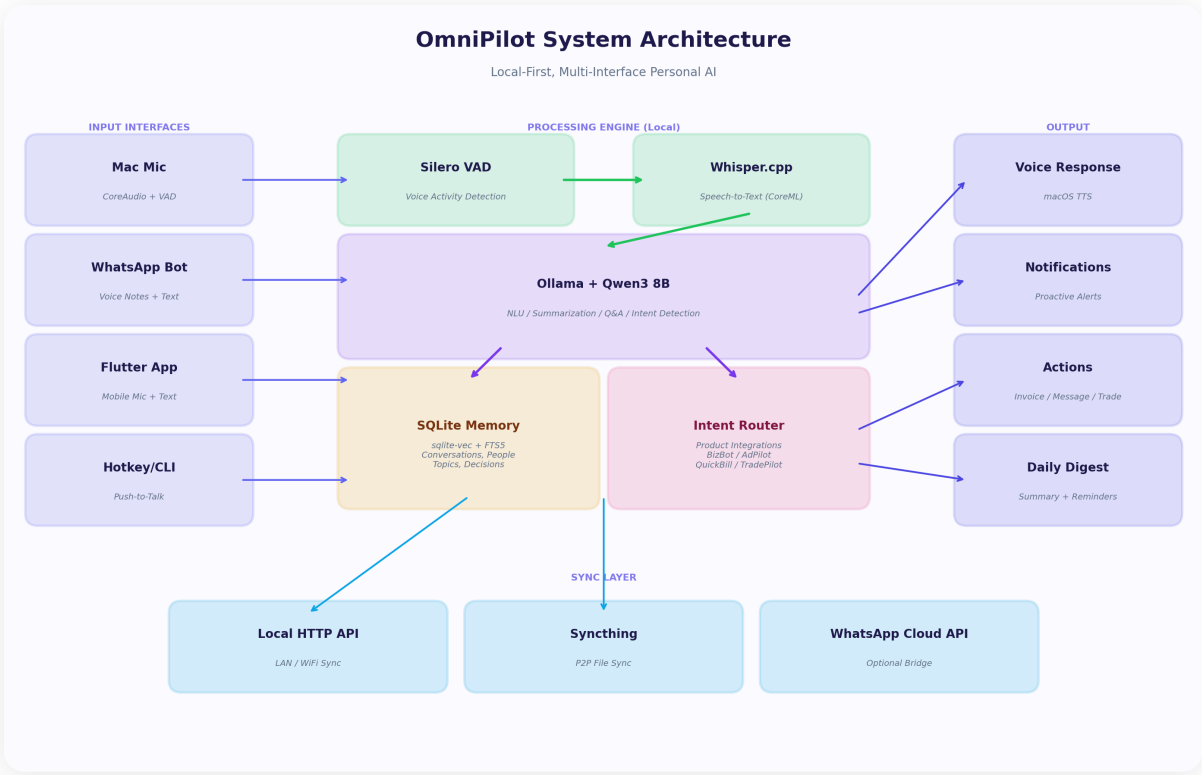
Without FTS5, searching would mean reading every single memory one by one (slow). With FTS5, it builds an **index** (like a book's index at the back) so lookups are instant.

Our Memory Schema (How Data Is Organized)



Each "memory" stores: **what was said** (content), **when** (timestamp), **who** (participants), **about what** (topics), and **how important** (score). This lets you search by any dimension.

How It All Connects



The Full Journey of Your Voice

1. You say: "Let's launch the campaign next Monday"
2. [CoreAudio] captures the sound waves as numbers
3. [VAD] checks: "Is this speech or silence?" → YES, it's speech!
4. [Whisper] converts the audio to text:
"Let's launch the campaign next Monday"
5. [SQLite] stores this with timestamp & metadata
6. [LLM] extracts: Topics=[campaign, launch], Decisions=[launch Monday]
7. Later, you ask: "When did I say we'd launch?"
8. [FTS5] finds the memory → [LLM] answers:
"You said to launch the campaign next Monday, on April 16."

What Each Technology Brings

TECHNOLOGY	ROLE	HUMAN EQUIVALENT	SIZE
Swift	The app itself	Your assistant's body	2.7MB
CoreAudio	Mic capture	Ears	Built into Mac
VAD	Speech detection	Attention filter	<1KB
Whisper	Speech → text	Transcriptionist	465MB model
SQLite + FTS5	Memory storage	Filing cabinet	Built into Mac
Ollama + Llama	Understanding	Brain	2GB model

Key AI Concepts You Now Own

1. On-Device AI vs Cloud AI

THE MOST IMPORTANT CONCEPT

Most AI products (ChatGPT, Siri, Alexa) run in the **cloud** — your data is sent to someone else's computer, processed there, and results sent back. OmniPilot runs **entirely on your Mac**. No internet needed. No data shared. This is called "**on-device AI**" or "**local AI**" or "**edge AI**."

Why this matters for a personal assistant: OmniPilot hears everything you say. If that data went to a cloud, someone could potentially access all your conversations. Local = private by design.

2. Tokens & Parameters

LLM VOCABULARY

Token: A piece of a word. "OmniPilot" might be 3 tokens: "Omni" + "Pil" + "ot". LLMs think in tokens, not words. When we say "20-30 tokens/second," that's roughly 15-22 words/second.

Parameters: The "knowledge settings" in an AI model. Llama 3.2 has 3 billion of these. More parameters = more knowledge and reasoning ability, but also more RAM and slower speed.

3. Embeddings & Vector Search

FUTURE UPGRADE (COMING IN SPRINT 2)

Embedding: Converting text into a list of numbers (a "vector") that captures its *meaning*, not just the words. "Happy" and "joyful" would have very similar number lists, even though they're different words.

Vector search: Finding similar meanings, not just matching keywords. When you search "budget discussion," vector search also finds memories about "financial planning" and "cost estimates" even if they don't contain the word "budget."

Currently we use **keyword search** (FTS5). Vector search will make OmniPilot much smarter at finding relevant memories.

4. RAG (Retrieval-Augmented Generation)

WHAT OMNIPLOT ACTUALLY DOES WHEN YOU ASK A QUESTION

RAG is a technique where instead of asking the LLM to answer from its training data (which knows nothing about YOUR conversations), we first **retrieve relevant memories** from the database, then **give them to the LLM as context**, and ask it to answer based on those.

Without RAG: "What did I discuss about budget?" → LLM guesses (wrong).

With RAG: "Here are 5 memories mentioning budget. Now answer the question." → LLM gives an accurate, sourced answer.

What's Next: The Road Ahead



Technology Upgrades Coming

UPGRADE	WHAT IT ADDS	WHEN
Silero VAD	Neural network VAD — much more accurate than energy-based	Sprint 2
sqlite-vec	Vector search — find memories by meaning, not just keywords	Sprint 2
Speaker diarization	Know WHO said what in a meeting (not just what was said)	Sprint 2
WhatsApp bot	Send voice notes to OmniPilot via WhatsApp	Sprint 3
Flutter mobile app	OmniPilot on your phone with on-device transcription	Sprint 4
BizBot integration	"Reply to that customer on WhatsApp" via voice	Sprint 5
Bigger LLM	Upgrade to 8B model when disk space allows — smarter answers	When needed

Glossary of Terms

TERM	WHAT IT MEANS
LLM	Large Language Model — AI that understands and generates text
STT	Speech-to-Text — converting spoken words to written text
VAD	Voice Activity Detection — detecting when someone is talking
RAG	Retrieval-Augmented Generation — giving the AI relevant context before asking

TERM	WHAT IT MEANS
FTS5	Full-Text Search — fast keyword search over all memories
Embedding	Converting text to numbers that capture meaning
Vector	A list of numbers representing something (text, image, audio)
Model	A file containing everything an AI has learned during training
Parameters	The adjustable settings inside an AI model (billions of them)
Token	A piece of a word — how LLMs process text
CoreML	Apple's AI acceleration framework (uses M1/M2 chip's Neural Engine)
Ollama	Tool to run LLMs locally on your Mac
Pipeline	A chain of processing steps where output of one feeds into the next
ONNX	A format for AI models that runs on any device
SQLite	A database stored as a single file — no server needed

OmniPilot

You don't just own an app.
You own the *entire AI stack* — ears, brain, and memory —
running privately on your Mac.

No subscriptions. No cloud. No one listening except you.



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